

Project Progress & Architecture Report

Project: Video Synopsis AI (AI-Powered YouTube Video Analyzer)
Prepared By: Intern (M. Sravanthi Suma)
Status: Development Phase Active (Backend Verified & Connected)
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1. Project Executive Overview

Video Synopsis AI is an enterprise-grade AI-powered web application built during the internship at Symbiosys Technologies. The system automates the summarization, categorization, and analysis of YouTube videos. It downloads video transcripts (with multiple extraction fallbacks, including OpenAI Whisper speech-to-text models), processes them via advanced LLM analysis (GPT-4o-mini), performs emotional sentiment analysis, and formats the output into interactive timelines, actionable item lists, and downloadable Word (DOCX) or PDF summaries.

2. Full Technology Stack

The system is designed with a decoupled, modular client-server architecture:

Component	Technologies Used	Key Responsibilities
Frontend (Web Client)	React 18, TypeScript, Vite, Tailwind CSS, React Router DOM, Axios, Lucide Icons	Responsive Dark-Mode UI, Interactive Dashboard, Video Analytics panels, PDF generation, Local Storage sandboxing
Backend (REST API)	FastAPI (Python 3.10), Uvicorn, Motor (Async MongoDB Driver), Pydantic v2, PyJWT, passlib[bcrypt]	RESTful controllers, JWT User authentication, payload validation, MongoDB CRUD endpoints, Word (DOCX) compilation
Data Storage	MongoDB Atlas (Cloud-native Cluster), NoSQL collections (users, synopses)	Asynchronous document stores with relational user-to-synopsis schema mapping
AI / Processing Pipeline	OpenAI API (GPT-4o-mini), OpenAI Whisper (Speech-to-Text), YouTube Transcript API, yt-dlp	Transcription scraping, audio extraction fallback, dynamic summarization, keyword indexing, sentiment scoring

3. Configurations & API Keys Used

For security and operation, all credentials and API endpoints are loaded dynamically from environment files (.env). These are verified as active and running in your local workspace:

Configuration Variable	Current Active Value	Description / Notes
MONGODB_URL	mongodb+srv://Sravanthi634:Sravanthi@cluster0.d9pyvsc.mongodb.net/?appName=Cluster0	Live cloud MongoDB Atlas Cluster connection string. Successfully pinged and verified active.
OPENAI_API_KEY	sk-proj-VM35fJRR7ok8J3...po4G4WJwMxOToAVJX4A	Used for Whisper Audio Transcription fallback and GPT-4o-mini summarization.
YOUTUBE_API_KEY	AlzaSyBzTeZSlzqla1_r9dk-6ldWsXagC8rm5o	Used for fetching video metadata (thumbnails, title, channel name, release date).
JWT_SECRET	44efb3c8f87a8bdfa7322e7d7008cfcfda5e656d2db9697193b2a3faef96c641	Secret cryptographic key used to sign and verify user authentication tokens.
Backend API URL	http://127.0.0.1:5000	Server host address. Port 5000 is directly exposed to and consumed by the React app.

WARNING: Avoid committing the `.env` file containing live API keys to public version control systems (like GitHub). Keep it in `.gitignore` to protect the billing resources.

4. Detailed Progress: Completed Work (Pin-to-Pin)

Frontend Implementation (100% Complete)

- **10 Completed Pages:** Fully styled with modern custom CSS and tailwind animations.
- • *LandingPage*: Showcases Whisper Pipelines, AI models, and internship scope.
- • *LoginPage & RegisterPage*: Secure portals supporting credential validation and dynamic Admin/User role toggles.
- • *ForgotPasswordPage*: Handles simulated email reset link submissions.
- • *DashboardPage*: Grid displaying saved summaries count, user statistics, search filters, and keyword tags.
- • *GenerateSynopsisPage*: Interface validating YouTube links, offering language selections, summary length models (Short/Med/Long), and live progress logs.
- • *SynopsisViewerPage*: Displays structured results with interactive chapter lists, circular tonality gauge, insights, and checklists.
- • *ProfilePage*: User details, active sessions list, name editing, and security key changes.
- • *AdminPage*: Statistics charts for summaries generated, user account table (with role promotion/deletion), and detailed API error lists.
- **14 Custom Components:** Fully functional reusable modules (Navbar, Footer, Sidebar, ProtectedRoute, AdminRoute, Button, Input, Loader, Alert, VideoCard, SummaryCard, HistoryTable, DownloadButtons, ProgressTracker).
- **Dual-Mode Service Layers:** All frontend services (authService, synopsisService, adminService) are engineered with an IS_DEMO_MODE toggle. Setting this to false immediately redirects the UI to communicate with the FastAPI server.

Backend Implementation (95% Complete)

- **MongoDB Atlas Integration:** Fully configured motor databases. Startup lifespan routine successfully establishes database connection.
- **User authentication:** Registration and login routes fully functional. Implemented password hashing (bcrypt) and access token generation (PyJWT). Profile updating and security password changes are fully handled.
- **Multi-Stage AI Pipeline:** Core video processing logic in SynopsisService runs seamlessly:
 - Extracts video ID from YouTube URL.
 - Pulls metadata (channelName, publishDate, thumbnail) from YouTube URL.
 - Tries fetching native transcript, falls back to yt-dlp caption tracks, then falls back to OpenAI Whisper transcribing, and lastly uses custom procedural metadata fallback to prevent OpenAI API quota exceptions.
 - Sends formatted transcript with timestamps to GPT-4o-mini using JSON schema validation to produce structured reports.
 - Validates schema against Pydantic models and inserts into MongoDB.
- **Document (DOCX) Export:** Completed server-side document creation (using python-docx and lxml) allowing users to export their summaries directly as Word files.
- **Platform Diagnostics:** Health check route (/api/health) checks API status and queries MongoDB connection directly.
- **API Swagger Docs:** Fully active at <http://127.0.0.1:5000/docs> allowing interactive REST testing.

5. Remaining Work (Next Steps)

- **Connect Frontend to Backend:** Toggle IS_DEMO_MODE from true to false in the frontend services to switch the frontend app from sandbox mode to live database interactions.
- **Admin Logging Expansion:** Populate the error log dashboard with real-time error exceptions (currently returns empty placeholder metrics in the backend admin endpoints).

- **Deployment & Hosting:** Build the frontend assets (npm run build) and host them (e.g. Vercel/Netlify), and deploy the backend container (e.g. Render/AWS/Railway).
- **Public Rate Limiting:** Add API rate limiters (e.g. SlowAPI) to avoid exploitation of OpenAI tokens by public requests.